🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 5 · STANDARD 6.GR.1

Determine the Area of Parallelograms and Rhombuses

Lesson 5-1 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes

Level 3 Enrichment

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can find the area of a parallelogram using base × height.

Language Objective: I can explain how I found the area using the words base, height, area, and formula.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Area of Parallelograms · Slanted side · Parallelogram · Parallel · Base · Height · Perpendicular · Area · Composite figure · Formula

👆 *Tap any word to see what it means and a picture.*

Level 3: try the blanks from memory first, then check the bank.

1. The space inside a parallelogram, found with $A = b \times h$, is the ________.

2. The leaning side of a parallelogram, which is NOT the height, is the ________.

3. A quadrilateral with two pairs of parallel sides is a ________.

4. Two sides that stay the same distance apart and never meet are ________.

5. The side of a parallelogram you measure the height from at a 90° angle is the

   ______________.

6. The perpendicular distance between the base and the opposite side is the

   ______________.

7. Two lines that meet at a 90° square corner are ______________.

8. The amount of flat space inside a two-dimensional figure is its

   ______________.

9. A shape made of two or more basic shapes combined is a

   ______________.

10. A math rule written with symbols, like A = b × h, is a

    ______________.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

How many tiles does the architect need?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- [ ] **Area of Parallelograms** — The space inside a parallelogram. Formula: $A = b \times h$, using the perpendicular height.
  Área de paralelogramos — El espacio dentro de un paralelogramo. Fórmula: $A = b \times h$, usando la altura perpendicular.

- [ ] **Slanted side** — A side that leans instead of going straight up and down. It is not the perpendicular height.
  Lado inclinado — Un lado que está inclinado en vez de subir y bajar verticalmente. No es la altura perpendicular.

- [ ] **Parallelogram** — A four-sided shape with two pairs of parallel sides.
  Paralelogramo — Una figura de cuatro lados con dos pares de lados paralelos.

- [ ] **Parallel** — Two lines or sides that stay the same distance apart and never meet.
  Paralelas — Dos líneas o lados que se mantienen a la misma distancia y nunca se cruzan.

- [ ] **Base** — Any side of a parallelogram can be the base. Once you pick a base, the height must be measured perpendicular (at a 90° angle) to it.
  Base — Cualquier lado de un paralelogramo puede ser la base. Una vez que eliges la base, la altura debe medirse perpendicular (en un ángulo de 90°) a ella.

Say it first: Say your idea to a partner or quietly to yourself before you write.
Di tu idea a un compañero o en voz baja antes de escribir.

The courtyard's area is ____ sq ft because A = ____ x ____ . She needs ____ tiles because ____ / 2 = ____.
Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: We use the height because it is perpendicular to the base.

Evidence: Students connect 'height' to the perpendicular (straight-up) distance from base to top, and recognize the slanted side is longer and would overstate the area.

Reasoning: This evidence connects the definition of area of parallelograms to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The courtyard's area is ____ sq ft because A = ____ x ____ . She needs ____ tiles because ____ / 2 = ____.**
- **My answer is ____.**
  Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **I know because ____.**
  Lo sé porque ____.
- **So ____.**
  Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
  Mi afirmación es ____.
- **My math evidence is ____.**
  Mi evidencia matemática es ____.
- **This proves ____ because ____.**
  Esto demuestra ____ porque ____.

4. Check Your Explanation

- [ ] I answered the question.
  Respondí la pregunta.
- [ ] I used math evidence: numbers, an equation, a model, or a comparison.
  Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- [ ] I explained how the evidence proves my answer.
  Expliqué cómo la evidencia demuestra mi respuesta.
- [ ] I used at least two math words.
  Usé por lo menos dos palabras matemáticas.
- [ ] I reread complete sentences and punctuation.
  Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Area of Parallelograms
2. **Notes 2:** Slanted side
3. **Notes 3:** Parallelogram
4. **Notes 4:** Parallel
5. **Notes 5:** Base
6. **Notes 6:** Height
7. **Notes 7:** Perpendicular
8. **Notes 8:** Area
9. **Notes 9:** Composite figure
10. **Notes 10:** Formula
11. **Watch for this mistake:** *The most common mistake in Area of Parallelograms is multiplying the base by the slanted side instead of the perpendicular height. For a parallelogram with a base of 16 ft, a perpendicular height of 9 ft, and a slanted side of 11 ft, students compute 16 × 11 = 176 square feet instead of the correct 16 × 9 = 144 square feet. The height must be the straight-up (perpendicular) distance between the base and its opposite side, not the length of the slanted side, even though the slanted side is often the number that 'looks' like it belongs in the formula.*
12. **Try It 1:** A. 40 sq ft — *Area = base x height = 8 x 5 = 40 sq ft. The 7 ft slanted side is not used; only the perpendicular height belongs in the formula.*
13. **Try It 2:** A. Equal: both are 10 x 6 = 60 sq ft because area depends on base and height, not the slant — *Area = base x height. Both have base 10 ft and height 6 ft, so both areas are 10 x 6 = 60 sq ft. Tilting the shape changes the slanted side but not the perpendicular height, so the area stays the same.*